

Physics 7B Midterm 2 Review Session Solutions

Electric field, Gauss's law, potential, and capacitors

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1 Electric Charge and Electric Field

Example 1.1: Why not just make it a semi-circle?

Suppose you have a uniformly charged unclosed circle of total charge Q and radius 1, centered at $(0, 0)$, with the arc above $y = 1/\sqrt{2}$ removed. Find the electric field at the origin.

Solution

The missing arc is the top arc from $\theta = \pi/4$ to $\theta = 3\pi/4$. Since the full circle would give zero field at the origin, the field from the remaining charge distribution is the negative of the field from the missing arc.

The remaining arc length is

$$L_{\text{rem}} = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2},$$

so the linear charge density is

$$\lambda = \frac{Q}{L_{\text{rem}}} = \frac{2Q}{3\pi}$$

For an element $dq = \lambda d\theta$ on a unit circle, the field magnitude at the origin is $dE = \frac{1}{4\pi\epsilon_0} dq$, and its direction is toward the center:

$$d\vec{E} = -\frac{1}{4\pi\epsilon_0} dq (\cos\theta \hat{x} + \sin\theta \hat{y})$$

Thus the field of the missing arc is

$$\vec{E}_{\text{miss}} = -\frac{1}{4\pi\epsilon_0} \lambda \int_{\pi/4}^{3\pi/4} (\cos\theta \hat{x} + \sin\theta \hat{y}) d\theta$$

The x -component vanishes by symmetry. For the y -component,

$$E_{\text{miss},y} = -\frac{1}{4\pi\epsilon_0} \lambda \int_{\pi/4}^{3\pi/4} \sin\theta d\theta = -\frac{1}{4\pi\epsilon_0} \lambda \left[-\cos\theta \right]_{\pi/4}^{3\pi/4} = -\frac{1}{4\pi\epsilon_0} \lambda \sqrt{2}$$

So

$$\vec{E}_{\text{rem}} = -\vec{E}_{\text{miss}} = \frac{1}{4\pi\epsilon_0} \lambda \sqrt{2} \hat{y}$$

Substitute $\lambda = 2Q/(3\pi)$:

$$\boxed{\vec{E}(0) = \frac{\sqrt{2}Q}{6\pi^2\epsilon_0} \hat{y}}$$

The field points upward.

Example 1.2: Bob's issue with E&M

Paul, of charge $-Q$, sits at the origin. Bob has charge $+Q$. Initially there is a uniform external field

$$\vec{E}_{\text{ext}} = E \hat{x}, \quad E > 0$$

Bob wants the net field at the point halfway between him and Paul to be zero.

- (a) Where should Bob position himself?
- (b) Suppose the external field suddenly changes to $\vec{E}_{\text{ext}} = E \hat{y}$. How much work does Bob have to do to cancel this new field?

Solution

(a) Let Bob be placed at $x = a > 0$. Then the midpoint is at $x = a/2$.

At that midpoint, the field from Paul ($-Q$ at the origin) points toward the origin, i.e. in the $-\hat{i}$ direction, with magnitude

$$E_P = \frac{1}{4\pi\epsilon_0} \frac{Q}{(a/2)^2} = \frac{4Q}{4\pi\epsilon_0 a^2}$$

The field from Bob ($+Q$ at $x = a$) at the midpoint also points in the $-\hat{x}$ direction, with the same magnitude. Therefore the two charges together give

$$\vec{E}_{\text{charges}} = -\frac{8Q}{4\pi\epsilon_0 a^2} \hat{x}$$

To cancel the external field $E \hat{x}$, we require

$$E - \frac{8Q}{4\pi\epsilon_0 a^2} = 0$$

Hence

$$a^2 = \frac{8Q}{4\pi\epsilon_0 E} \quad \Rightarrow \quad a = \sqrt{\frac{8Q}{4\pi\epsilon_0 E}} = \sqrt{\frac{2Q}{\pi\epsilon_0 E}}$$

So Bob should sit on the $+x$ -axis at

$$(x, y) = \left(\sqrt{\frac{2Q}{\pi\epsilon_0 E}}, 0 \right)$$

(b) After the external field rotates to $E \hat{j}$, Bob must move so that the midpoint field from the two charges points in the $-\hat{j}$ direction. By the same reasoning, Bob should now sit directly above Paul at the same separation

$$(x, y) = \left(0, \sqrt{\frac{2Q}{\pi\epsilon_0 E}} \right)$$

The Bob–Paul interaction energy is unchanged, because their separation remains a . So only Bob's potential energy in the *new* external field changes.

With the new field $\vec{E} = E \hat{y}$, the electric potential is $V(y) = -Ey + \text{const}$, so Bob's potential energy is

$$U = qV = -QEy + \text{const}$$

Initially $y_i = 0$, and finally $y_f = a$, so

$$\Delta U = -QEa$$

Thus the work Bob must do quasistatically is

$$W_{\text{Bob}} = \Delta U = -QEa = -QE\sqrt{\frac{2Q}{\pi\epsilon_0 E}}$$

This is negative, meaning the external field actually does the work for him; Bob does not need to supply energy.

Practice 1.3: Charged tube on its axis

An uncapped tube of radius R and length L carries uniform surface charge density σ .

- Explain why you cannot use Gauss' law to determine the electric field at a point on the x -axis.
- Determine the field created by a very thin slice of the tube at any point on the x -axis, taking the origin at the center of the thin ring.
- Determine the field created by the whole charged tube at any point on the x -axis, taking the origin at the right end of the cylinder.

Solution

(a) Gauss' law is always true, but it is only directly useful when symmetry lets us choose a Gaussian surface on which $\vec{E}$ is either constant and normal to the surface, or tangent everywhere.

For a finite uncapped cylinder, the field on a coaxial Gaussian cylinder is *not* uniform, and there is also nonzero flux through the end caps. So symmetry is not strong enough to extract E from Gauss' law alone.

(b) A very thin slice of thickness dx' is a ring of radius R carrying charge

$$dq = \sigma(2\pi R dx')$$

At a point on the axis a distance x from the ring center, transverse components cancel by symmetry, so only the x -component remains:

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{dq}{x^2 + R^2} \cos \alpha, \quad \cos \alpha = \frac{x}{\sqrt{x^2 + R^2}}$$

Therefore

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{dq x}{(x^2 + R^2)^{3/2}} = \frac{1}{4\pi\epsilon_0} \frac{2\pi R\sigma x dx'}{(x^2 + R^2)^{3/2}}$$

So

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2\pi R\sigma x dx'}{(x^2 + R^2)^{3/2}} \hat{x}$$

(c) Let the right end of the tube be at $x = 0$, so the tube occupies $-L \leq x' \leq 0$. At a field point x on the axis, a ring at position x' contributes

$$dE_x = \frac{1}{4\pi\epsilon_0} \frac{2\pi R\sigma(x - x') dx'}{[(x - x')^2 + R^2]^{3/2}}$$

Thus

$$E_x = \frac{1}{4\pi\epsilon_0} 2\pi R\sigma \int_{-L}^0 \frac{x - x'}{[(x - x')^2 + R^2]^{3/2}} dx'$$

Let $u = x - x'$. Then $du = -dx'$, and the limits become $u = x + L$ to $u = x$, giving

$$E_x = \frac{1}{4\pi\epsilon_0} 2\pi R\sigma \int_x^{x+L} \frac{u}{(u^2 + R^2)^{3/2}} du$$

Using

$$\int \frac{u du}{(u^2 + R^2)^{3/2}} = -\frac{1}{\sqrt{u^2 + R^2}},$$

we get

$$E_x = \frac{1}{4\pi\epsilon_0} 2\pi R\sigma \left(\frac{1}{\sqrt{x^2 + R^2}} - \frac{1}{\sqrt{(x+L)^2 + R^2}} \right)$$

Therefore

$$\vec{E}(x) = \frac{\sigma R}{2\epsilon_0} \left(\frac{1}{\sqrt{x^2 + R^2}} - \frac{1}{\sqrt{(x+L)^2 + R^2}} \right) \hat{x}$$

This expression is valid for any point on the axis.

2 Gauss's Law

Example 2.1: Gauss–shmauss, as long as I'm right

An infinitely long cylinder of radius R_1 has volume charge density

$$\rho(r) = \frac{Q_0 r^2}{2\pi}$$

Around it is an infinitely thin conducting cylindrical shell of radius R_2 .

- Find the electric field for $r < R_2$.
- What charge per unit length on the outer shell makes the field outside vanish?

Solution

By cylindrical symmetry, for a Gaussian cylinder of radius r and length ℓ ,

$$E(2\pi r\ell) = \frac{Q_{\text{enc}}}{\epsilon_0}$$

(a) Region $0 < r < R_1$. The enclosed charge is

$$Q_{\text{enc}}(r) = \int_0^r \rho(r') (2\pi r' \ell) dr' = \int_0^r \frac{Q_0 r'^2}{2\pi} (2\pi r' \ell) dr' = Q_0 \ell \int_0^r r'^3 dr' = \frac{Q_0 \ell r^4}{4}$$

Hence

$$E(2\pi r\ell) = \frac{Q_0\ell r^4}{4\epsilon_0} \implies \boxed{\vec{E}(r) = \frac{Q_0 r^3}{8\pi\epsilon_0} \hat{\mathbf{r}}, \quad 0 < r < R_1}$$

Region $R_1 < r < R_2$. Now the entire inner cylinder is enclosed, so

$$Q_{\text{enc}} = \frac{Q_0\ell R_1^4}{4}$$

Thus

$$E(2\pi r\ell) = \frac{Q_0\ell R_1^4}{4\epsilon_0} \implies \boxed{\vec{E}(r) = \frac{Q_0 R_1^4}{8\pi\epsilon_0 r} \hat{\mathbf{r}}, \quad R_1 < r < R_2}$$

So altogether, for $r < R_2$,

$$\boxed{\vec{E}(r) = \begin{cases} \frac{Q_0 r^3}{8\pi\epsilon_0} \hat{\mathbf{r}}, & 0 < r < R_1 \\ \frac{Q_0 R_1^4}{8\pi\epsilon_0 r} \hat{\mathbf{r}}, & R_1 < r < R_2 \end{cases}}$$

(b) To make the field outside vanish, the net enclosed charge per unit length must be zero. The inner cylinder carries charge per unit length

$$\lambda_{\text{inner}} = \frac{1}{\ell} Q_{\text{inner}} = \frac{Q_0 R_1^4}{4}$$

Therefore the outer shell must carry

$$\boxed{\lambda_{\text{outer}} = -\frac{Q_0 R_1^4}{4}}$$

3 Electric Potential

Example 3.1: A potentially fun problem?

- (a) What is the potential at the origin of Problem 1.1 if $V(\infty) = 0$? What is the speed of an electron (charge $-e$, mass m_e) at the origin if it is released from rest at ∞ ?
- (b) Neglecting the outer cylinder in Problem 2.1, what is the potential outside a distance r from the center of the inner cylinder? Define $V(R_2) = 0$. Qualitatively, what would the potential be inside the cylinder if the inner cylinder were instead a conductor?

Solution

(a) In Problem 1.1 every charge element lies a distance 1 from the origin, so

$$V(0) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{1} = \frac{1}{4\pi\epsilon_0} Q$$

Thus

$$V(0) = \frac{Q}{4\pi\epsilon_0}$$

An electron released from rest at infinity has initial total energy 0 (since $V(\infty) = 0$ and $K_i = 0$). At the origin,

$$K_f + U_f = 0, \quad U_f = (-e)V(0).$$

So

$$K_f = eV(0) = \frac{eQ}{4\pi\epsilon_0}$$

Hence

$$\frac{1}{2}m_e v^2 = \frac{eQ}{4\pi\epsilon_0} \quad \Rightarrow \quad v = \sqrt{\frac{eQ}{2\pi\epsilon_0 m_e}}$$

(b) For $r > R_1$, the field of the inner cylinder alone is

$$E(r) = \frac{Q_0 R_1^4}{8\pi\epsilon_0 r}$$

Using $V(R_2) = 0$,

$$V(r) - V(R_2) = - \int_{R_2}^r E(r') dr' = - \frac{Q_0 R_1^4}{8\pi\epsilon_0} \int_{R_2}^r \frac{dr'}{r'}$$

Therefore

$$V(r) = \frac{Q_0 R_1^4}{8\pi\epsilon_0} \ln \frac{R_2}{r}, \quad r \geq R_1$$

If the inner cylinder were a conductor instead, then the field inside it would be zero, so the potential would be constant throughout the conductor and equal to its surface value:

$$V(r) = V(R_1) \quad \text{for } r < R_1$$

In other words, the conductor interior would be an equipotential region.

Practice 3.2: CO₂ molecule

The CO₂ molecule is modeled as a linear charge distribution

$$\{A(-q), O(+2q), B(-q)\},$$

with $AO = OB = a$.

(a) Determine the electric potential $V(r, \theta)$ for $r \gg a$.

(b) Explain how to calculate (E_r, E_θ) from this potential.

Solution

Take the molecular axis as the polar axis, with the $+2q$ charge at the origin and the two $-q$ charges at $z = \pm a$.

The exact potential is

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{2q}{r} - \frac{q}{r_A} - \frac{q}{r_B} \right)$$

For $r \gg a$, use the standard expansion

$$\frac{1}{r_{A,B}} \approx \frac{1}{r} \left[1 \mp \frac{a \cos \theta}{r} + \frac{a^2}{2r^2} (3 \cos^2 \theta - 1) \right]$$

Adding the two outer charges,

$$\frac{1}{r_A} + \frac{1}{r_B} \approx \frac{2}{r} + \frac{a^2}{r^3} (3 \cos^2 \theta - 1)$$

Therefore

$$V(r, \theta) \approx \frac{1}{4\pi\epsilon_0} \left[\frac{2q}{r} - q \left(\frac{2}{r} + \frac{a^2}{r^3} (3 \cos^2 \theta - 1) \right) \right]$$

The monopole and dipole terms cancel, leaving

$$V(r, \theta) \approx -\frac{qa^2}{4\pi\epsilon_0 r^3} (3 \cos^2 \theta - 1), \quad r \gg a$$

(b) Once $V(r, \theta)$ is known, the electric field follows from

$$\vec{E} = -\nabla V$$

In spherical coordinates (with no ϕ dependence here),

$$E_r = -\frac{\partial V}{\partial r}, \quad E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta}$$

That is the required procedure.

4 Capacitors

Example 4.1: A Capacity to Learn

A parallel-plate capacitor of area A and plate separation d is filled by two dielectric slabs stacked along the field direction, each of thickness $d/2$, with dielectric constants $\kappa_1 > \kappa_2$. The top plate is negatively charged.

- Sketch the electric field as a function of distance from the negative plate if the plates carry charges $\pm Q$.
- Sketch the potential $V(x)$ and determine the total potential difference.

(c) Find the capacitance.

Solution

Let x measure downward from the negative plate. The electric displacement is the same in both dielectric layers:

$$D = \frac{Q}{A}$$

Since $D = \varepsilon E = \kappa \varepsilon_0 E$,

$$E_1 = \frac{Q}{\kappa_1 \varepsilon_0 A}, \quad E_2 = \frac{Q}{\kappa_2 \varepsilon_0 A}$$

The field points from the positive plate to the negative plate, i.e. upward, so in x -coordinates,

$$\vec{E} = \begin{cases} -\frac{Q}{\kappa_1 \varepsilon_0 A} \hat{x}, & 0 < x < d/2 \\ -\frac{Q}{\kappa_2 \varepsilon_0 A} \hat{x}, & d/2 < x < d \end{cases}$$

So the field magnitude is piecewise constant, with the larger value in the lower- κ material.

Potential. Since $E_x = -dV/dx$,

$$\frac{dV}{dx} = \begin{cases} \frac{Q}{\kappa_1 \varepsilon_0 A}, & 0 < x < d/2 \\ \frac{Q}{\kappa_2 \varepsilon_0 A}, & d/2 < x < d \end{cases}$$

Choosing $V(0) = 0$ for convenience,

$$V(x) = \begin{cases} \frac{Qx}{\kappa_1 \varepsilon_0 A}, & 0 \leq x \leq d/2 \\ \frac{Qd}{2\kappa_1 \varepsilon_0 A} + \frac{Q(x - d/2)}{\kappa_2 \varepsilon_0 A}, & d/2 \leq x \leq d \end{cases}$$

Hence the total potential difference is

$$\Delta V = V(d) - V(0) = \frac{Qd}{2\varepsilon_0 A} \left(\frac{1}{\kappa_1} + \frac{1}{\kappa_2} \right)$$

Therefore the capacitance is

$$C = \frac{Q}{\Delta V} = \frac{2\varepsilon_0 A}{d \left(\frac{1}{\kappa_1} + \frac{1}{\kappa_2} \right)} = \boxed{\frac{2\kappa_1 \kappa_2 \varepsilon_0 A}{d(\kappa_1 + \kappa_2)}}$$

Example 4.2: A Capacity to Learn More

A parallel-plate capacitor of area A and plate separation d is filled by two dielectrics placed side-by-side, each covering half the plate area.

(a) Find the capacitance.

(b) If the plates carry charges $\pm q$, find the electric field in the two regions.

Solution

This is two capacitors in parallel, each with area $A/2$ and separation d . Their capacitances are

$$C_1 = \frac{\kappa_1 \varepsilon_0 (A/2)}{d}, \quad C_2 = \frac{\kappa_2 \varepsilon_0 (A/2)}{d}$$

So

$$C = C_1 + C_2 = \frac{\varepsilon_0 A}{2d} (\kappa_1 + \kappa_2)$$

For capacitors in parallel, the voltage is the same across both regions. Since $q = CV$,

$$V = \frac{q}{C} = \frac{2qd}{\varepsilon_0 A (\kappa_1 + \kappa_2)}$$

Thus the field magnitude in each region is

$$E_1 = E_2 = \frac{V}{d} = \frac{2q}{\varepsilon_0 A (\kappa_1 + \kappa_2)}$$

The direction is from the positive plate to the negative plate.

Practice 4.3: Coaxial capacitor & dielectric

Two coaxial conducting cylindrical shells of length L and radii $R_1 < R_2$ carry charges $+Q$ and $-Q$ respectively, with $L \gg R_2$.

- Find the electric field.
- Find the magnitude of the voltage between the shells.
- Determine the capacitance.
- Determine the capacitance when the gap is filled with two radial dielectric layers: K_1 from R_1 to $R_1 + d$, and K_2 from $R_1 + d$ to R_2 .

Solution

(a) By Gauss' law, for $R_1 < r < R_2$,

$$E(2\pi r L) = \frac{Q}{\varepsilon_0} \quad \Rightarrow \quad \vec{E}(r) = \frac{Q}{2\pi \varepsilon_0 L r} \hat{r}$$

Inside the conductors, $E = 0$. For a long ideal capacitor, the outside field is also negligible (and in the infinite-length idealization, zero).

(b) The voltage magnitude is

$$|V| = \int_{R_1}^{R_2} E(r) dr = \frac{Q}{2\pi \varepsilon_0 L} \int_{R_1}^{R_2} \frac{dr}{r}$$

Hence

$$|V| = \frac{Q}{2\pi\epsilon_0 L} \ln \frac{R_2}{R_1}$$

(c) Therefore

$$C = \frac{Q}{|V|} = \frac{2\pi\epsilon_0 L}{\ln(R_2/R_1)}$$

(d) In the dielectric-filled case,

$$E_1(r) = \frac{Q}{2\pi K_1 \epsilon_0 L r}, \quad R_1 < r < R_1 + d$$

$$E_2(r) = \frac{Q}{2\pi K_2 \epsilon_0 L r}, \quad R_1 + d < r < R_2$$

So the total voltage is

$$|V| = \frac{Q}{2\pi\epsilon_0 L} \left[\frac{1}{K_1} \ln \frac{R_1 + d}{R_1} + \frac{1}{K_2} \ln \frac{R_2}{R_1 + d} \right]$$

Thus

$$C = \frac{2\pi\epsilon_0 L}{\frac{1}{K_1} \ln \frac{R_1 + d}{R_1} + \frac{1}{K_2} \ln \frac{R_2}{R_1 + d}}$$

Practice 4.4: Coaxial cylindrical shells in oil

Two long coaxial metallic cylindrical shells of inner radius a and outer radius b are placed vertically in oil of density ρ and dielectric constant κ . A constant voltage V is maintained across the shells. The oil rises to height h between the shells.

- Determine the capacitance when there is no oil.
- Determine the capacitance when the oil has risen to height h .
- Calculate ΔU_{batt} and ΔU_{capa} .
- Determine the sign of the total energy change and explain physically where the energy goes.

Solution

(a) With no oil present, the capacitor is just an air/vacuum coaxial capacitor of length L :

$$C_0 = \frac{2\pi\epsilon_0 L}{\ln(b/a)}$$

(b) The portion of height h filled with oil acts like a coaxial capacitor of length h and dielectric constant κ , while the remaining height $L - h$ acts like a vacuum capacitor. These two parts are

in parallel (same voltage across each), so

$$C(h) = \frac{2\pi\epsilon_0(L-h)}{\ln(b/a)} + \frac{2\pi\kappa\epsilon_0 h}{\ln(b/a)}$$

Hence

$$C(h) = \frac{2\pi\epsilon_0}{\ln(b/a)} [L + (\kappa - 1)h]$$

So the capacitance change is

$$\Delta C = C(h) - C_0 = \frac{2\pi\epsilon_0(\kappa - 1)h}{\ln(b/a)}$$

(c) Because the voltage is held fixed, the battery supplies extra charge

$$\Delta Q = V \Delta C$$

The work done *by* the battery is $W_{\text{batt}} = V \Delta Q = V^2 \Delta C$. Therefore the battery's own potential energy changes by the negative of this:

$$\Delta U_{\text{batt}} = -V^2 \Delta C = -\frac{2\pi\epsilon_0(\kappa - 1)hV^2}{\ln(b/a)}$$

The capacitor energy is

$$U_{\text{capa}} = \frac{1}{2} CV^2,$$

so

$$\Delta U_{\text{capa}} = \frac{1}{2} V^2 \Delta C = \frac{\pi\epsilon_0(\kappa - 1)hV^2}{\ln(b/a)}$$

(d) The total change is

$$\Delta U_{\text{tot}} = \Delta U_{\text{batt}} + \Delta U_{\text{capa}} = -\frac{1}{2} V^2 \Delta C = -\frac{\pi\epsilon_0(\kappa - 1)hV^2}{\ln(b/a)}$$

Thus

$$\Delta U_{\text{tot}} < 0$$

The battery-plus-capacitor system loses energy. That released energy goes into pulling the oil upward against gravity (and possibly some dissipation), which is why the oil rises into the capacitor.

5 Energy and Motion in Electric Fields

Practice 5.1: Electron released along the axis of a charged disk

An electron of mass m and charge $-e$, initially at rest, is released from infinity along the symmetry axis of a uniformly charged disk of radius R and surface charge density σ . Find its kinetic energy at a distance z from the center of the disk.

Solution

For a uniformly charged disk, the potential on its axis (taking $V(\infty) = 0$) is

$$V(z) = \frac{\sigma}{2\epsilon_0} \left(\sqrt{z^2 + R^2} - z \right), \quad z \geq 0$$

The electron starts from rest at infinity, so its initial total energy is zero. At position z ,

$$K + U = 0, \quad U = (-e)V(z)$$

Therefore

$$K = eV(z)$$

Hence

$$K(z) = \frac{e\sigma}{2\epsilon_0} \left(\sqrt{z^2 + R^2} - z \right)$$

Practice 5.2: Electric field & potential from a model atom

A model atom has nuclear charge $q > 0$ at the origin and a surrounding negative charge distribution such that the potential is

$$V(r) = \frac{q}{4\pi\epsilon_0 r} e^{-r/a}$$

- Determine the direction and magnitude of the electric field.
- Calculate the electric flux through a sphere of radius r .
- Calculate the total enclosed charge $Q_{\text{tot}}(r)$ and its limit as $r \rightarrow \infty$.
- Determine the negative charge distribution $\rho(r)$.

Solution

(a) Since the potential is spherically symmetric,

$$\vec{E}(r) = -\frac{dV}{dr} \hat{r}$$

Differentiate:

$$\frac{d}{dr} \left(\frac{e^{-r/a}}{r} \right) = -e^{-r/a} \left(\frac{1}{ar} + \frac{1}{r^2} \right)$$

Therefore

$$\vec{E}(r) = \frac{q}{4\pi\epsilon_0} e^{-r/a} \left(\frac{1}{r^2} + \frac{1}{ar} \right) \hat{\mathbf{r}}$$

The field points radially outward.

(b) The flux through a sphere of radius r is

$$\Phi_E = E(r)(4\pi r^2)$$

Substituting the field,

$$\Phi_E(r) = \frac{q}{\epsilon_0} e^{-r/a} \left(1 + \frac{r}{a} \right)$$

(c) By Gauss' law,

$$Q_{\text{tot}}(r) = \epsilon_0 \Phi_E(r)$$

So

$$Q_{\text{tot}}(r) = q e^{-r/a} \left(1 + \frac{r}{a} \right)$$

As $r \rightarrow \infty$,

$$\lim_{r \rightarrow \infty} Q_{\text{tot}}(r) = 0$$

So the total charge of the whole atom is zero, as expected for a neutral atom.

(d) For $r > 0$, the enclosed charge changes only because of the surrounding negative cloud, so

$$\frac{dQ_{\text{tot}}}{dr} = 4\pi r^2 \rho(r)$$

Differentiate:

$$\frac{d}{dr} \left[q e^{-r/a} \left(1 + \frac{r}{a} \right) \right] = -\frac{qr}{a^2} e^{-r/a}$$

Hence

$$4\pi r^2 \rho(r) = -\frac{qr}{a^2} e^{-r/a},$$

so

$$\rho(r) = -\frac{q}{4\pi a^2 r} e^{-r/a}, \quad r > 0$$

This is the negative cloud density. If one wants the *total* charge density everywhere, one must add the nuclear point charge:

$$\rho_{\text{total}}(\mathbf{r}) = q \delta^{(3)}(\mathbf{r}) - \frac{q}{4\pi a^2 r} e^{-r/a}$$